

20

Conformal Mappings

EXERCISES 20.1

Complex Functions as Mappings

1. For $w = \frac{1}{z}$, $u = \frac{x}{x^2 + y^2}$ and $v = \frac{-y}{x^2 + y^2}$. If $y = x$, $u = \frac{1}{2} \frac{1}{x}$, $v = -\frac{1}{2} \frac{1}{x}$, and so $v = -u$. The image is the line $v = -u$ (with the origin $(0, 0)$ excluded.)
2. If $y = 1$, $u = \frac{x}{x^2 + 1}$ and $v = \frac{-1}{x^2 + 1}$. It follows that $u^2 + v^2 = \frac{1}{x^2 + 1} = -v$ and so $u^2 + (v + \frac{1}{2})^2 = (\frac{1}{2})^2$. This is a circle with radius $r = \frac{1}{2}$ and center at $(0, -\frac{1}{2}) = -\frac{1}{2}i$. The circle can also be described by $|w + \frac{1}{2}i| = \frac{1}{2}$.
3. For $w = z^2$, $u = x^2 - y^2$ and $v = 2xy$. If $xy = 1$, $v = 2$ and so the hyperbola $xy = 1$ is mapped onto the line $v = 2$.
4. If $x^2 - y^2 = 4$, $u = 4$ and so the hyperbola $x^2 - y^2 = 4$ is mapped onto the vertical line $u = 4$.
5. For $w = \text{Ln } z$, $u = \log_e |z|$ and $v = \text{Arg } z$. The semi-circle $|z| = 1$, $y > 0$ may also be described by $r = 1$, $0 < \theta < \pi$. Therefore $u = 0$ and $0 < v < \pi$. The image is therefore the open line segment from $z = 0$ to $z = \pi i$.
6. If $\theta = \pi/4$, then $v = \theta = \pi/4$. In addition $u = \log_e r$ will vary from $-\infty$ to ∞ . The image is therefore the horizontal line $v = \pi/4$.
7. For $w = z^{1/2} = (re^{i\theta})^{1/2} = r^{1/2}e^{i\theta/2}$ and $\theta = \theta_0$, $w = \sqrt{r}e^{i\theta_0/2}$. Therefore $\text{Arg } w = \theta_0/2$ and so the image is the ray $\theta = \theta_0/2$.
8. If $r = 2$ and $0 \leq \theta \leq \frac{\pi}{2}$, $w = \sqrt{2}e^{i\theta/2}$. Therefore $|w| = \sqrt{2}$ and $0 \leq \text{Arg } w \leq \pi/4$. This image is a circular arc.
9. For $w = e^z$, $u = e^x \cos y$ and $v = e^x \sin y$. Therefore if $e^x \cos y = 1$, $u = 1$. The curve $e^x \cos y = 1$ is mapped into the line $u = 1$. Since $v = \frac{\sin y}{\cos y} = \tan y$, v varies from $-\infty$ to ∞ and the image is the line $u = 1$.
10. If $w = z + \frac{1}{z}$ and $z = e^{it}$, $w = e^{it} + e^{-it} = 2 \cos t$. Therefore $u = 2 \cos t$ and $v = 0$ and so the image is the closed interval $[-2, 2]$ on the u -axis.
11. The first quadrant may be described by $r > 0$, $0 < \theta < \pi/2$. If $w = 1/z$ and $z = re^{i\theta}$, $w = \frac{1}{r}e^{-i\theta}$. Therefore $\text{Arg } w = -\theta$ and so $-\pi/2 < \text{Arg } w < 0$. The image is therefore the fourth quadrant.
12. For $w = \frac{1}{z}$, $u = \frac{x}{x^2 + y^2}$ and $v = \frac{-y}{x^2 + y^2}$. The line $y = 0$ is mapped to the line $v = 0$, and, from Problem 2, the line $y = 1$ is mapped onto the circle $|w + \frac{1}{2}i| = \frac{1}{2}$. Since $f(\frac{1}{2}i) = -2i$, the region $0 \leq y \leq 1$ is mapped onto the points in the half-plane $v \leq 0$ which are on or outside the circle $|w + \frac{1}{2}i| = \frac{1}{2}$. (The image does not include the point $w = 0$.)

20.1 Complex Functions as Mappings

13. Since $w = e^{x+iy} = e^x e^{iy}$ and $\pi/4 \leq y \leq \pi/2$, $\pi/4 \leq \text{Arg } w \leq \pi/2$ and $|w| = e^x$. The image is therefore the angular wedge defined by $\pi/4 \leq \text{Arg } w \leq \pi/2$.
14. Since $w = e^{x+iy} = e^x e^{iy}$ and $0 \leq x \leq 1$, $0 \leq y \leq \pi$, we have $|w| = e^x$ and $\text{Arg } w = y$. Therefore $1 \leq |w| \leq e$ and $0 \leq \text{Arg } w \leq \pi$. These inequalities define a semi-angular region in the w -plane.
15. The mapping $w = z + 4i$ is a translation which maps the circle $|z| = 1$ to a circle of radius $r = 1$ and with center $w = 4i$. This circle may be described by $|w - 4i| = 1$.
16. If $w = 2z - 1$ and $|z| = 1$, then, since $z = \frac{w+1}{2}$, $\left| \frac{w+1}{2} \right| = 1$ or $|w+1| = 2$. The image is a circle with center at $w = -1$ and with radius $r = 2$.
17. The mapping $w = iz$ is a rotation through 90° since $i = e^{i\pi/2}$. Therefore the strip $0 \leq y \leq 1$ is rotated through 90° and so the strip $-1 \leq u \leq 0$ is the image in the w -plane.
18. Since $w = (1+i)z = \sqrt{2}e^{i\pi/4}z$, the mapping is the composite of a rotation through 45° and a magnification by $\alpha = \sqrt{2}$. The image of the first quadrant is therefore the angular wedge $\pi/4 \leq \text{Arg } w \leq 3\pi/4$.
19. The power function $w = z^3$ changes the opening of the wedge $0 \leq \text{Arg } z \leq \pi/4$ by a factor of 3. Therefore the image region is $0 \leq \text{Arg } w \leq 3\pi/4$.
20. The power function $w = z^{1/2}$ changes the opening of the wedge $0 \leq \text{Arg } z \leq \pi/4$ by a factor of $1/2$. Therefore the image region is $0 \leq \text{Arg } w \leq \pi/8$.
21. We first let $z_1 = z - i$ to map the region $1 \leq y \leq 4$ to the region $0 \leq y_1 \leq 3$. We then let $w = e^{-i\pi/2}z_1$ to rotate this strip through -90° . Therefore $w = -i(z - i) = -iz - 1$ maps $1 \leq y \leq 4$ to the strip $0 \leq u \leq 3$.
22. The mapping $w = z - i$ lowers the strip $1 \leq y \leq 4$ one unit so that the image is $0 \leq v \leq 3$ in the w -plane.
23. We first let $z_1 = z - 1$ to map the disk $|z - 1| \leq 1$ to the disk $|z_1| \leq 1$. We then use the magnification $w = 2z_1$ to obtain $|w| \leq 2$ as the image. The composite of these two mappings is $w = 2(z - 1)$.
24. The mapping $w = iz$ will rotate the strip $-1 \leq x \leq 1$ through 90° so that the strip $-1 \leq v \leq 1$ results.
25. We first use $z_1 = e^{-i\pi/4}z$ to rotate the wedge $\pi/4 \leq \text{Arg } z \leq \pi/2$ to the wedge $0 \leq \text{Arg } z_1 \leq \pi/4$. The power function $w = z_1^4$ then changes the opening of this wedge by a factor of 4 so that the strip $0 \leq \text{Arg } w \leq \pi$ results. The composite of these two mappings is $w = (e^{-i\pi/4}z)^4 = e^{-\pi i}z^4 = -z^4$.
26. The magnification $z_1 = \frac{\pi}{4}z$ maps the strip $0 \leq y \leq 4$ to the strip $0 \leq y_1 \leq \pi$. By Example 1, Section 20.1, $w = e^{z_1}$ maps this strip onto the upper half-plane $v \geq 0$. The composite of these two mappings is $w = e^{\frac{\pi}{4}z}$.
27. By Example 1, Section 20.1, $z_1 = e^z$ maps the strip $0 \leq y \leq \pi$ onto the upper half-plane $y_1 \geq 0$, or $0 \leq \text{Arg } z_1 \leq \pi$. The power function $w = z_1^{3/2}$ changes the opening of this wedge by a factor of $3/2$ so the wedge $0 \leq \text{Arg } w \leq 3\pi/2$ results. The composite of these two mappings is $w = (e^z)^{3/2} = e^{3z/2}$.
28. The power function $z_1 = z^{2/3}$ maps the wedge $0 \leq \text{Arg } z \leq 3\pi/2$ to the upper half-plane $0 \leq \text{Arg } z_1 \leq \pi$. We then let $w = e^{-i\pi/2}z_1 + 2 = -iz_1 + 2$ to rotate the upper half-plane through -90° and then translate 2 units to the right. Therefore the composite function is $w = -iz^{2/3} + 2$ and the image region is the half-plane $u \geq 2$.
29. We may obtain the image region by first rotating R through 180° and then raising the resulting region one unit in the vertical direction. Therefore $w = e^{i\pi}z + i = -z + i$.
30. The mapping $z_1 = -(z - \pi i)$ lowers R by π units in the vertical direction and then rotates the resulting region through 180° . The image region R_1 is upper half-plane $y_1 \geq 0$. By Example 1, Section 20.1, $w = \text{Ln } z_1$ maps R_1 onto the strip $0 \leq v \leq \pi$. The composite of these two mappings is $w = \text{Ln}(\pi i - z)$.

31. (a) Letting $z = x + iy$ and noting that in this case $x^2 + y^2 = R^2$, the transformation $w = z + k^2/z$ becomes

$$\begin{aligned} w &= x + iy + \frac{k^2}{x + iy} = x + iy + \frac{k^2}{x^2 + y^2} (x - iy) \\ &= x + iy + \frac{k^2}{R^2} (x - iy) = \left(1 + \frac{k^2}{R^2}\right)x + i\left(1 - \frac{k^2}{R^2}\right)y. \end{aligned}$$

We identify $u = (1 + k^2/R^2)x$ and $v = (1 - k^2/R^2)y$. Then

$$\frac{u^2}{\left(1 + \frac{k^2}{R^2}\right)^2} = x^2 \quad \text{and} \quad \frac{v^2}{\left(1 - \frac{k^2}{R^2}\right)^2} = y^2, \quad k \neq R,$$

so that

$$\frac{u^2}{\left(1 + \frac{k^2}{R^2}\right)^2} + \frac{v^2}{\left(1 - \frac{k^2}{R^2}\right)^2} = x^2 + y^2 = R^2, \quad k \neq R.$$

- (b) When $R = k$ the circle $z = ke^{it}$ is transformed into $w = z + k^2/z = ke^{it} + ke^{-it} = 2k \cos t + 0i$. Thus, the circle $z = ke^{it}$ is transformed into the closed interval $[-2k, 2k]$ on the u -axis.
- (c) Letting $w = z + k^2/z$ we have

$$\frac{w - 2k}{w + 2k} = \frac{z + \frac{k^2}{z} - 2k}{z + \frac{k^2}{z} + 2k} = \frac{z^2 + k^2 - 2kz}{z^2 + k^2 + 2kz} = \frac{(z - k)^2}{(z + k)^2} = \left(\frac{z - k}{z + k}\right)^2.$$

EXERCISES 20.2

Conformal Mappings

1. Since $f'(z) = 3(z^2 - 1)$, f is conformal at all points except $z = \pm 1$.
2. Since $f'(z) = -\sin z$, f is conformal at all points except $z = \pm n\pi$.
3. $f'(z) = 1 + e^z$ and $1 + e^z = 0$ for $z = \pm i \pm 2n\pi i$. Therefore f is conformal except for $z = \pi i \pm 2n\pi i$.
4. $f(z) = z + \operatorname{Ln} z + 1$ is analytic for all z except $z = y \leq 0$. But $f'(z) = 1 + 1/z$ and $1 + 1/z = 0$ for $z = -1$. Therefore $f(z)$ is conformal at all points except those on the branch cut $y \leq 0$.
5. $f(z) = (z^2 - 1)^{1/2} = e^{\frac{1}{2}\operatorname{Ln}(z^2 - 1)}$ is analytic for all z outside the interval $[-1, 1]$ on the real axis. This follows from the fact that $z^2 - 1 = (x^2 - y^2 - 1) + 2xyi$ and so we must exclude values of z for which $v = 2xy = 0$ and $u = x^2 - y^2 - 1 \leq 0$. Therefore $y = 0$ and $x^2 \leq 1$. $f'(z) = z/(z^2 - 1)^{1/2}$ is non-zero outside this interval. Therefore f is conformal except for $z = x$, $-1 \leq x \leq 1$.
6. The function $f(z) = \pi i - \frac{1}{2}[\operatorname{Ln}(z + 1) + \operatorname{Ln}(z - 1)]$ is analytic except on the branch cut $x - 1 \leq 0$ or $x \leq 1$, and

$$f'(z) = -\frac{1}{2} \left(\frac{1}{z + 1} + \frac{1}{z - 1} \right) = -\frac{z}{z^2 - 1}$$

is non-zero for $z \neq 0, \pm 1$. Therefore f is conformal except for $z = x$, $x \leq 1$.

20.2 Conformal Mappings

7. $f(z) = \cos z = \sin(\pi/2 - z)$ is the composite of $z_1 = \pi/2 - z$ and $w = \sin z$. The strip $0 \leq x \leq \pi$ is mapped onto the strip $-\pi/2 \leq x \leq \pi/2$ by $z_1 = \pi/2 - z$ and $w = \sin z$ maps this strip onto the region shown in Figure 20.11(b). The horizontal segment $z(t) = t + ib$, $0 < t < \pi$ is first mapped to the horizontal segment $z_i(t) = (\pi/2 - t) - ib$, $0 < t < \pi$. This latter segment is mapped onto the lower or upper portion of the ellipse

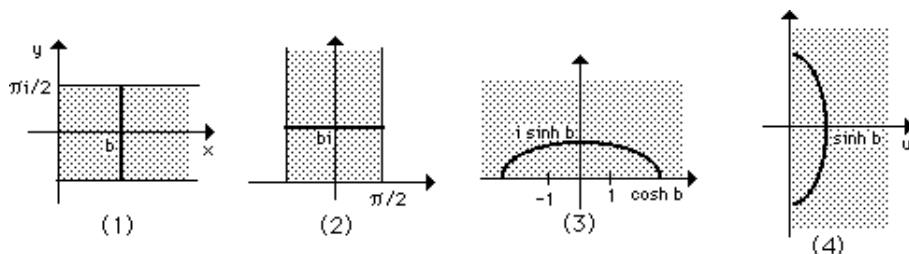
$$\frac{u^2}{\cosh^2 b} + \frac{v^2}{\sinh^2 b} = 1$$

according to whether $b > 0$ or $b < 0$. See Figure 20.11.

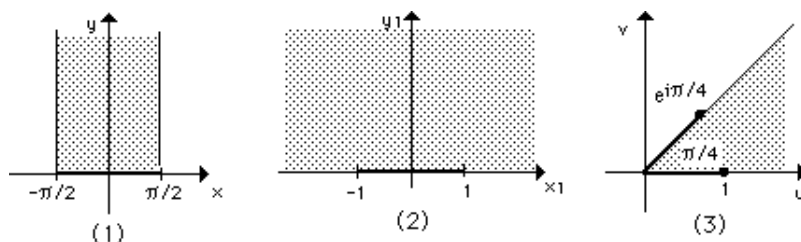
8. $f(z) = \sinh z = -i \sin(iz)$ is the composite of $z_1 = iz$, $z_2 = \sin z_1$, and $w = -iz_2$. The strip $-\pi/2 \leq y \leq \pi/2$, $x \geq 0$ is rotated through 90° so that the image is the strip $-\pi/2 \leq x \leq \pi/2$, $y \geq 0$. This latter strip is mapped to the upper half-plane $y_2 \geq 0$ by $z_2 = \sin z_1$, and $w = -iz_2$ rotates this upper half-plane through -90° . The final image region is the half-plane $u \geq 0$. A vertical line segment in the original strip is mapped to the right hand side of the ellipse

$$\frac{u^2}{\sinh^2 b} + \frac{v^2}{\cosh^2 b} = 1.$$

See the figures below.

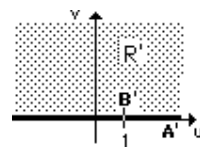


9. $f(z) = (\sin z)^{1/4}$ is the composite of $z_1 = \sin z$ and $w = z_1^{1/4}$. The region $-\pi/2 \leq x \leq \pi/2$, $y \geq 0$ is mapped to the upper half-plane $y_2 \geq 0$ by $z_1 = \sin z$ (See Example 2) and the power function $w = z_1^{1/4}$ maps this upper half-plane to the angular wedge $0 \leq \text{Arg } w \leq \pi/4$. The real interval $[-\pi/2, \pi/2]$ is first mapped to $[-1, 1]$ and then to the union of the line segments from $e^{i\pi/4}$ to 0 and 0 to 1. See the figures below.

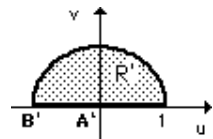


10. From Example 3, $u = (r + 1/r) \cos \theta$ and $v = (r - 1/r) \sin \theta$. If $r = 1$, $u = 2 \cos \theta$ and $v = 0$. Therefore the image of the circle $|z| = 1$ is the real interval $[-2, 2]$. If $0 < r < 1$, $u^2/a^2 + v^2/b^2 = 1$ where $a = r + 1/r > 2$ and $b = 1/r - r$. The resulting ellipse together with $[-2, 2]$ fill up the entire w -plane. Therefore the image of the region $|z| \leq 1$ is the entire w -plane. If $z = x$ and $-1 \leq x \leq 1$, $w = x + 1/x$. Therefore w is real and, from an analysis of the graph of $w = x + 1/x$ for $-1 \leq x \leq 1$, $|w| \geq 2$. Therefore the segment $[-1, 1]$ (with 0 excluded) is mapped on those points $w = u$ on the u -axis for which $|u| \geq 2$.

11. Using H-4 with $a = 2$, $w = \cos(\pi z/2)$ maps R onto the target region R' . The image of AB is shown in the figure.

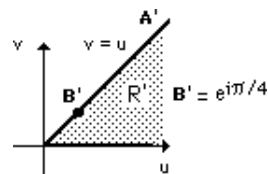


12. Using C-3 $w = e^z$ maps R onto the target region R' . The image of AB is shown in the figure.



13. Using H-5, $z_1 = \left(\frac{1+z}{1-z}\right)^2$ maps R onto the upper half-plane $y_1 \geq 0$, and $w = z_1^{1/4}$ maps this half-plane onto the target region R' . The image of AB is shown in the figure, and

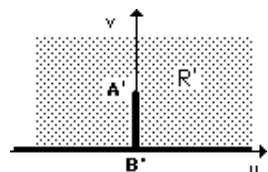
$$w = \left(\frac{1+z}{1-z}\right)^{1/2}.$$



14. Using H-1, $z_1 = i \frac{1-z}{1+z}$ maps R onto the upper half-plane $y_1 \geq 0$, and using M-4 with $a = 1$, $w = (z_1^2 - 1)^{1/2}$ maps this upper half-plane onto the target region R' . Therefore

$$w = \left[-\left(\frac{1-z}{1+z}\right)^2 - 1 \right]^{1/2}$$

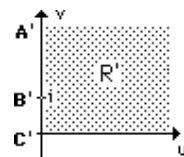
and the image of AB is shown in the figure.



15. Using H-6, $z_1 = \frac{e^{\pi/z} + e^{-\pi/z}}{e^{\pi/z} - e^{-\pi/z}}$ maps R onto the upper half-plane $y_1 \geq 0$, and $w = z_1^{1/2}$ maps this half-plane onto the target region R' . Therefore

$$w = \left(\frac{e^{\pi/z} + e^{-\pi/z}}{e^{\pi/z} - e^{-\pi/z}} \right)^{1/2}$$

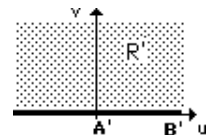
and the image of AB is shown in the figure.



16. We can translate R to the origin, magnify by 2, and then use H-1 to reach the target region R' . Therefore $z_1 = 2(z - \frac{1}{2})$, $w = i \frac{1-z}{1+z}$ and so

$$w = i \frac{1 - (2z - 1)}{1 + (2z - 1)} = i \frac{1 - z}{z}.$$

The image of AB is shown in the figure.



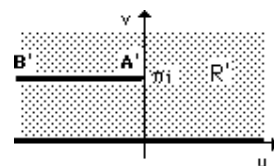
17. $z_1 = \text{Ln } z$ maps R onto the horizontal strip $0 \leq y_1 \leq \pi$, and, to prepare this strip for mapping by the sine function, we let $z_2 = -iz_1 - \pi/2$ to obtain the vertical strip $-\pi/2 \leq x \leq \pi/2$. Finally $w = \sin z_2$ maps this vertical strip onto the target region R' . Therefore

$$w = \sin \left(-i \text{Ln } z - \frac{\pi}{2} \right).$$

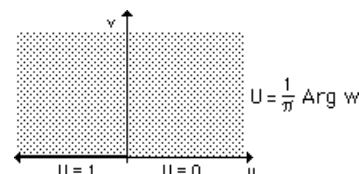
The image of AB is the real interval $(-\infty, -1]$.

20.2 Conformal Mappings

18. Using E-9, $z_1 = \cosh z$ maps R onto the upper half-plane $y_1 \geq 0$. Using M-7, $w = z_1 + \operatorname{Ln} z_1 + 1$ maps this half-plane onto the target region R' . Therefore $w = \cosh z + \operatorname{Ln}(\cosh z) + 1$ and the image of AB is shown in the figure.

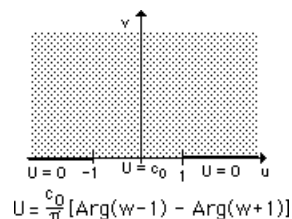


In Exercises 19-22, we find a conformal mapping $w = f(z)$ that maps the given region R onto the upper half-plane $v \geq 0$ and transfers the boundary conditions so that the resulting Dirichlet problem is as shown in the figure.



19. $f(z) = z^4$ and so $u = U(f(z)) = \frac{1}{\pi} \operatorname{Arg} z^4$. The solution may also be written as $u(r, \theta) = 4\theta/\pi$.
20. $f(z) = \left(\frac{1+z}{1-z}\right)^2$, using H-5, and so $u = U(f(z)) = \frac{1}{\pi} \operatorname{Arg} \left(\frac{1+z}{1-z}\right)^2$.
21. $f(z) = i \frac{1-z}{1+z}$, using H-1, and so $u = U(f(z)) = \frac{1}{\pi} \operatorname{Arg} \left(\frac{1-z}{1+z}\right)$. The solution may also be written as $u(x, y) = \frac{1}{\pi} \tan^{-1} \left(\frac{1-x^2-y^2}{2y}\right)$.
22. $f(z) = \frac{1}{2} \left(z + \frac{1}{z}\right)$, using H-3 with $a = 1$, and so $u = U(f(z)) = \frac{1}{\pi} \operatorname{Arg} \frac{1}{2} \left(z + \frac{1}{z}\right)$. The solution may also be written as $u(x, y) = \frac{1}{\pi} \tan^{-1} \left[\frac{y}{x} \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1}\right]$.

In Exercises 23-26, we find a conformal mapping $w = f(z)$ that maps the given region R onto the upper half-plane $v \geq 0$ and transfers the boundary conditions so that the resulting Dirichlet problem is as shown in the figure.



23. $f(z) = z^2$ and $c_0 = 1$. Therefore $u = \frac{1}{\pi} [\operatorname{Arg}(z^2 - 1) - \operatorname{Arg}(z^2 + 1)]$.
24. The mapping $z_1 = z^2$ maps R onto the region R_1 defined by $y_1 \geq 0$, $|z_1| \geq 1$ and shown in H-3, and $w = \frac{1}{2} \left(z_1 + \frac{1}{z_1}\right)$ maps R_1 onto the upper half-plane $v \geq 0$. Letting $c_0 = 5$,
- $$u = \frac{5}{\pi} \left[\operatorname{Arg} \left(\frac{1}{2} \left[z^2 + \frac{1}{z^2} \right] - 1 \right) - \operatorname{Arg} \left(\frac{1}{2} \left[z^2 + \frac{1}{z^2} \right] + 1 \right) \right].$$
25. $f(z) = e^{\pi z}$, using H-2 with $a = 1$. Letting $c_0 = 10$,
- $$u = \frac{10}{\pi} [\operatorname{Arg}(e^{\pi z} - 1) - \operatorname{Arg}(e^{\pi z} + 1)].$$
26. $f(z) = \cos(\pi z/2)$ using H-4 with $a = 2$. Letting $c_0 = 4$,
- $$u = \frac{4}{\pi} [\operatorname{Arg}(\cos(\pi z/2) - 1) - \operatorname{Arg}(\cos(\pi z/2) + 1)].$$
27. (a) If $u = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} = 0$$

since ϕ is assumed to be biharmonic.

(b) If $g = u + iv$, then $\phi = \operatorname{Re}(\bar{z}g(z)) = xu + yv$.

$$\begin{aligned}\frac{\partial^2 \phi}{\partial x^2} &= 3 \frac{\partial u}{\partial x} + x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 v}{\partial x^2} \\ \frac{\partial^2 \phi}{\partial y^2} &= 2 \frac{\partial v}{\partial y} + x \frac{\partial^2 u}{\partial y^2} + y \frac{\partial^2 v}{\partial y^2}.\end{aligned}$$

Since u and v are harmonic and $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 2 \frac{\partial u}{\partial x} + 2 \frac{\partial v}{\partial y} = 4 \frac{\partial u}{\partial x}.$$

Now $u_1 = \frac{\partial u}{\partial x}$ is also harmonic and so $\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = 0$. But

$$\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = \frac{1}{4} \left[\frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} \right]$$

and so ϕ is biharmonic.

EXERCISES 20.3

Linear Fractional Transformations

- (a) For $T(z) = i/z$, $T(0) = \infty$, $T(1) = i$, and $T(\infty) = 0$.

(b) If $|z| = 1$, $|w| = |i/z| = 1/|z| = 1$. Therefore, the image of the circle $|z| = 1$ is the circle $|w| = 1$ in the w -plane. The circle $|z - 1| = 1$ passes through the pole at $z = 0$ and so the image is a line. Since $T(2) = \frac{1}{2}i$ and $T(1 + i) = \frac{1}{2} + \frac{1}{2}i$, the image is the line $v = \frac{1}{2}$.

(c) The disk $|z| \leq 1$ is mapped onto the disk $|w| \geq 1$.
- (a) For $T(z) = \frac{1}{z-1}$, $T(0) = -1$, $T(1) = \infty$, and $T(\infty) = 0$.

(b) The circle $|z| = 1$ passes through the pole at $z = 1$ and so the image is a line. Since $T(-1) = -\frac{1}{2}$ and $T(i) = -\frac{1}{2} - \frac{1}{2}i$, the image is the line $u = -\frac{1}{2}$. If $|z - 1| = 1$, $|w| = 1/|z - 1| = 1$ and the image is the circle $|w| = 1$ in the w -plane.

(c) Since $T(0) = -1$, the image of the disk $|z| \leq 1$ is the half-plane $u = -\frac{1}{2}$.
- (a) For $T(z) = \frac{z+1}{z-1}$, $T(0) = -1$, $T(1) = \infty$, and $T(\infty) = 1$.

(b) The circle $|z| = 1$ passes through the pole at $z = 1$ and so the image is a line. Since $T(-1) = 0$ and $T(i) = -i$, the image is the line $u = 0$. If $|z - 1| = 1$,

$$|w - 1| = \left| \frac{z+1}{z-1} - 1 \right| = \frac{2}{|z-1|} = 2$$

and so the image is the circle $|w - 1| = 2$ in the w -plane.

- (c) Since $T(0) = -1$, the image of the disk $|z| \leq 1$ is the half-plane $u \leq 0$.

20.3 Linear Fractional Transformations

4. (a) For $T(z) = \frac{z-i}{z}$, $T(0) = \infty$, $T(1) = 1-i$, and $T(\infty) = 1$.
 (b) $T(i) = 0$, $T(1) = 1-i$ and $T(-i) = 2$. The image of $|z| = 1$ is therefore a circle which passes through 0, $1-i$, and 2. This is the circle $|w-1| = 1$. The circle $|z-1| = 1$ passes through the pole at $z = 0$ and so the image is a line. Since $T(2) = 1 - \frac{1}{2}i$ and $T(1+i) = \frac{1}{2} - \frac{1}{2}i$, the image is the line $v = -\frac{1}{2}$.
 (c) $T(\frac{1}{2}) = 1-2i$ which is exterior to the circle $|w-1| = 1$. Therefore, the image of $|z| \leq 1$ is $|w-1| \geq 1$, the exterior of the circle $|w-1| = 1$ in the w -plane.

5. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} i & 1 \\ 1 & -1 \end{bmatrix} \right) \begin{bmatrix} 1 & 0 \\ i & -1 \end{bmatrix} = \begin{bmatrix} -1-i & 1 \\ -2 & -i \end{bmatrix}.$$

Therefore,

$$S^{-1}(T(z)) = \frac{(-1-i)z+1}{-2z-i} = \frac{(1+i)z-1}{2z+i} \quad \text{and} \quad S^{-1}(w) = \frac{-w-1}{-w+i} = \frac{w+1}{w-i}.$$

6. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \right) \begin{bmatrix} i & 0 \\ 1 & -2i \end{bmatrix} = \begin{bmatrix} -1+i & 2i \\ 2-i & -4i \end{bmatrix}.$$

Therefore,

$$S^{-1}(T(z)) = \frac{(-1+i)z+2i}{(2-i)z-4i} \quad \text{and} \quad S^{-1}(w) = \frac{w-1}{-w+2}.$$

7. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \right) \begin{bmatrix} 2 & -3 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & -3 \\ -1 & 0 \end{bmatrix}.$$

Therefore,

$$S^{-1}(T(z)) = \frac{-3}{-z} = \frac{3}{z} \quad \text{and} \quad S^{-1}(w) = \frac{-w+2}{-w+1} = \frac{w-2}{w-1}.$$

8. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} 2-i & 0 \\ 1 & -1-i \end{bmatrix} \right) \begin{bmatrix} 1 & -1+i \\ i & -2 \end{bmatrix} = \begin{bmatrix} -1-i & 2 \\ 2i & -3+i \end{bmatrix}.$$

Therefore,

$$S^{-1}(T(z)) = \frac{(-1-i)z+2}{2iz-3+i} \quad \text{and} \quad S^{-1}(w) = \frac{(-1-i)w}{-w+2-i} = \frac{(1+i)w}{w-2+i}.$$

9. $T(z) = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}$ maps z_1, z_2, z_3 to 0, 1, ∞ . Therefore, $T(z) = \frac{(z+1)(-2)}{(z-2)(1)} = -2 \frac{z+1}{z-2}$ maps $-1, 0, 2$ to 0, 1, ∞ .

10. $T(z) = \frac{(z-i)i}{(z+i)(-i)} = -\frac{z-i}{z+i}$ maps $i, 0, -i$ to 0, 1, ∞ .

11. $S(w) = \frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)}$ maps w_1, w_2, w_3 to $0, 1, \infty$ and so S' maps $0, 1, \infty$ to w_1, w_2, w_3 . Therefore,
 $z = \frac{(w - 0)(i - 2)}{(w - 2)(i - 0)}$ and so $w = \frac{2z}{z - 1 - 2i}$ maps $0, 1, \infty$ to $0, i, 2$.

12. As in Exercise 11, $z = \frac{(w - 1 - i)(-1 + i)}{(w - 1 + i)(-1 - i)}$ and, solving for w , $w = \frac{2z - 2}{(1 + i)z - 1 + i}$ maps $0, 1, \infty$ to $1 + i, 0, 1 - i$.

13. Using the cross-ratio formula (7),

$$\frac{(w - i)(1)}{w(1)} = \frac{(z + 1)(-1)}{(z - 1)(1)}$$

and so $w = \frac{i}{2} \frac{z - 1}{z}$ maps $-1, 0, 1$ to $i, \infty, 0$.

14. Using the cross-ratio formula (7),

$$\frac{(1)(-i - 1)}{(w - 1)(1)} = \frac{(z + 1)(-1)}{(z - 1)(1)}$$

and so $w = \frac{(2 + i)z - i}{z + 1}$ maps $-1, 0, 1$ to $\infty, -i, 1$.

15. Using the cross-ratio formula (7),

$$S(w) = \frac{(w + 1)(-3)}{(w - 3)(1)} = \frac{(z - 1)(2i)}{(z + i)(i - 1)} = T(z).$$

We can solve for w to obtain

$$w = 3 \frac{(1 + i)z + (1 - i)}{(-3 + 5i)z - 3 - 5i}.$$

Alternatively we can apply the matrix method to compute $w = S^{-1}(T(z))$.

16. Using the cross-ratio formula (7),

$$S(w) = \frac{(w - i)(-i - 1)}{(w - 1)(-2i)} = \frac{(z - 1)(2i)}{(z + i)(i - 1)} = T(z).$$

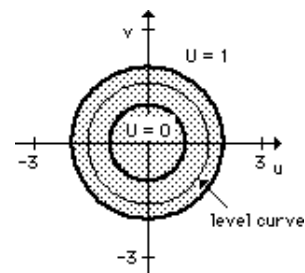
We can solve for w to obtain

$$w = \frac{(1 + 2i)z - i}{iz + 1 - 2i}.$$

Alternatively we can apply the matrix method to compute $w = S^{-1}(T(z))$.

17. From Example 2, $z = \frac{w + 2}{w - 1}$ maps the annular region $1 < |w| < 2$ onto the region R and the circle $|w| = 1$ corresponds to the line $x = -1/2$. Solving for w , $w = \frac{z + 2}{z - 1}$ maps R onto the annulus and the transferred boundary conditions are shown in the figure to the right. The solution to this new Dirichlet problem is $U = \log_e r / \log_e 2$ and so

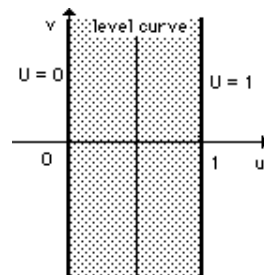
$$u = U(f(z)) = \frac{1}{\log_e 2} \log_e \left| \frac{z + 2}{z - 1} \right|$$



is the solution to the Dirichlet problem in Figure 20.37. The level curves are the images of the level curves of U , $|w| = r$ for $1 < r < 2$ under the mapping $z = \frac{w + 2}{w - 1}$. Since these circles do not pass through the pole at $w = 1$, the images are circles.

20.3 Linear Fractional Transformations

18. The mapping $T(z) = \frac{1}{2} \frac{z+1}{z}$ maps $-1, 1, 0$ to $0, 1, \infty$ and maps each of the two circles in R to lines since both circles pass through the pole at $z = 0$. Since $T(\frac{1}{2} + \frac{1}{2}i) = 1 - i$ and $T(1) = 1$, the circle $|z - \frac{1}{2}| = \frac{1}{2}$ is mapped onto the line $u = 1$. Likewise, the circle $|z + \frac{1}{2}| = \frac{1}{2}$ is mapped onto the line $u = 0$. The transferred boundary conditions are shown in the figure and $U(u, v) = u$ is the solution. The solution to the Dirichlet problem in Figure 20.38 is



$$u = U(T(z)) = \operatorname{Re} \left(\frac{1}{2} \frac{z+1}{z} \right) = \frac{1}{2} + \frac{1}{2} \frac{x}{x^2 + y^2}.$$

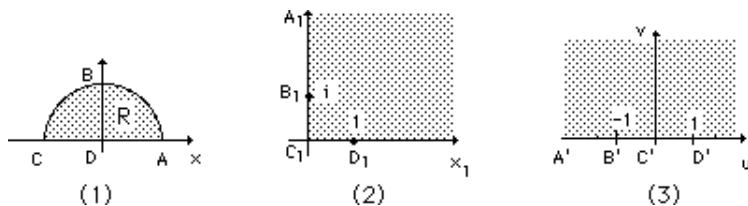
The level curves $u = c$ are the circles with centers on the x -axis which pass through the origin. The level curve $u = \frac{1}{2}$, however, is the vertical line $x = 0$.

19. The linear fractional transformation that sends $1, i, -i$ to $0, 1, -1$ satisfies the cross-ratio equation

$$\frac{(w-0)(2)}{(w+1)(1)} = \frac{(z-1)(2i)}{(z+i)(i-1)}.$$

Solving for w , $w = i \frac{1-z}{1+z} = T(z)$. Since $T(0) = i$, the image of the disk $|z| \leq 1$ is the upper half-plane $v \geq 0$.

20. The linear fractional transformation that sends $1, i, -1$ to $\infty, i, 0$ is $T(z) = \frac{1+z}{1-z}$ and so $f(z) = \left(\frac{1+z}{1-z} \right)^2$ maps $1, i, -1$ to $\infty, -1, 0$. The upper semi-circle is mapped by T to the positive imaginary axis, and the real interval $[-1, 1]$ is mapped to the positive real axis. Since $T\left(\frac{i}{2}\right) = \frac{3}{5} + \frac{4}{5}i$, the image of R under T is the first quadrant. $w = z_1^2$ doubles the size of the opening so that the image under f is the upper half-plane $v \geq 0$. See the figures below.



21. $T_2(T_1(z)) = \frac{a_2 T_1(z) + b_2}{c_2 T_1(z) + d_2} = \frac{a_2 \frac{a_1 z + b_1}{c_1 z + d_1} + b_2}{c_2 \frac{a_1 z + b_1}{c_1 z + d_1} + d_2} = \frac{a_1 a_2 z + a_2 b_1 + b_2 c_1 z + b_2 d_1}{a_1 c_2 z + b_1 c_2 + c_1 d_2 z + d_1 d_2} = \frac{(a_1 a_2 + c_1 b_2)z + (b_1 a_2 + d_1 b_2)}{(a_1 c_2 + c_1 d_2)z + (b_1 c_2 + d_1 d_2)}$
22. $|w - w_1| = \lambda |w - w_2| \implies (u - u_1)^2 + (v - v_1)^2 = \lambda^2 [(u - u_2)^2 + (v - v_2)^2]$

The latter equation may be put in the form

$$Au^2 + Bv^2 + Cu + Dv + F = 0$$

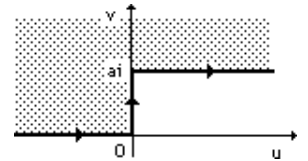
where $A = B = 1 - \lambda^2$. If $\lambda = 1$, the line $Cu + Dv + F = 0$ results. If $\lambda > 0$ and $\lambda \neq 1$, then the equation defines a circle.

EXERCISES 20.4

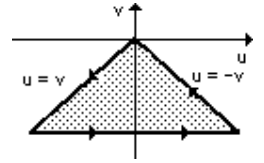
Schwarz-Christoffel Transformations

1. $\arg f'(t) = -\frac{1}{2}\text{Arg}(t-1) = \begin{cases} -\pi/2, & t < 1 \\ 0, & t > 1 \end{cases}$. Since $f(1) = 0$, the image is the first quadrant.
2. $\arg f'(t) = -\frac{1}{3}\text{Arg}(t+1) = \begin{cases} -\pi/3, & t < -1 \\ 0, & t > -1 \end{cases}$. In (2), $\alpha_1 = 2\pi/3$ and since $f(-1) = 0$, the image is the wedge $0 \leq \text{Arg } w \leq 2\pi/3$.

3. $\arg f'(t) = -\frac{1}{2}\text{Arg}(t+1) + \frac{1}{2}\text{Arg}(t-1) = \begin{cases} 0, & t < -1 \\ \pi/2, & -1 < t < 1 \\ 0, & t > 1 \end{cases}$
and $\alpha_1 = \pi/2$ and $\alpha_2 = 3\pi/2$. Since $f(-1) = 0$, the image of the upper half-plane is the region shown in the figure.



4. $\arg f'(t) = -\frac{1}{2}\text{Arg}(t+1) - \frac{3}{4}\text{Arg}(t-1) = \begin{cases} -5\pi/4, & t < -1 \\ -3\pi/4, & -1 < t < 1 \\ 0, & t > 1 \end{cases}$
and $\alpha_1 = \pi/2$ and $\alpha_2 = \pi/4$. Since $f(-1) = 0$, the image of the upper half-plane is the region shown in the figure.



5. Since $\alpha_1 = \alpha_2 = \alpha_3 = \pi/2$, $\alpha_i/\pi - 1 = -1/2$ and so $f'(z) = A(z+1)^{-1/2}z^{-1/2}(z-1)^{-1/2}$ for some constant A .
6. Since $\alpha_1 = \pi/3$ and $\alpha_2 = \pi/2$, $\alpha_1/\pi - 1 = -2/3$ and $\alpha_2/\pi - 1 = -1/2$ and so $f'(z) = A(z+1)^{-2/3}z^{-1/2}$ for some constant A .
7. Since $\alpha_1 = \alpha_2 = 2\pi/3$, $\alpha_i/\pi - 1 = -1/3$ and so $f'(z) = A(z+1)^{-1/3}z^{-1/3}$ for some constant A .
8. Since $\alpha_1 = 3\pi/2$ and $\alpha_2 = \pi/4$, $\alpha_1/\pi - 1 = 1/2$ and $\alpha_2/\pi - 1 = -3/4$. Therefore, $f'(z) = A(z+1)^{1/2}z^{-3/4}$ for some constant A .
9. Since $\alpha_1 = \alpha_2 = \pi/2$, $f'(z) = A(z+1)^{-1/2}(z-1)^{-1/2} = A/(z^2-1)^{1/2}$. Therefore, $f(z) = A \cosh^{-1} z + B$. But $f(-1) = \pi i$ and $f(1) = 0$. Since $\cosh^{-1} 1 = 0$, $B = 0$. Since $\cosh^{-1}(-1) = \pi i$, $\pi i = A(\pi i)$ and so $A = 1$. Hence $f(z) = \cosh^{-1} z$.
10. Since $\alpha_1 = 3\pi/2 = \alpha_2$, $f'(z) = A(z+1)^{1/2}(z-1)^{1/2} = A(z^2-1)^{1/2}$. Therefore,

$$f(z) = A \left[\frac{z(z^2-1)^{1/2}}{2} - \frac{1}{2} \text{Ln}(z + (z^2-1)^{1/2}) \right] + B.$$

but $f(-1) = -ai$ and $f(1) = ai$. It follows that

$$ai = f(1) = B, \quad -ai = f(-1) = A \left(-\frac{\pi i}{2} \right) + B$$

and so $B = ai$ and $A = 4a/\pi$. Therefore,

$$f(z) = \frac{4a}{\pi} \left[\frac{z(z^2-1)^{1/2}}{2} - \frac{1}{2} \text{Ln}(z + (z^2-1)^{1/2}) \right] + ai = \frac{4a}{\pi} \left[\frac{z(z^2-1)^{1/2}}{2} - \frac{1}{2} \cosh^{-1} z \right] + ai.$$

11. $f'(z) = A(z+1)^{(\alpha_1/\pi)-1}z^{(\alpha_2/\pi)-1}(z-1)^{(\alpha_3/\pi)-1}$ from (3). Since $f(-1) = \pi i$, $\alpha_1 \rightarrow \pi$ as $w_1 \rightarrow \infty$ in the horizontal direction. Likewise $\alpha_2 \rightarrow 0$ and $\alpha_3 \rightarrow \pi$. This suggests we examine $f'(z) = Az^{-1} = A/z$. Therefore,

20.4 Schwarz-Christoffel Transformations

$f(z) = A \ln z + B$. But $f(-1) = \pi i$ and $f(1) = 0$. It follows that $A = 1$ and $B = 0$ so that $f(z) = \ln z$. We verified in Example 1, Section 20.1 that $f(z) = \ln z$ maps the upper half-plane $y \geq 0$ to the horizontal strip $0 \leq v \leq \pi$.

12. From (3), $f'(z) = Az^{-3/4}(z-1)^{(\alpha_2/\pi)-1}$. But $\alpha_2 \rightarrow \pi$ as $\theta \rightarrow 0$. This suggests that we examine $f'(z) = Az^{-3/4}$. Therefore, $f(z) = A_1 z^{1/4} + B_1$. But $f(0) = 0$ and $f(1) = 1$ so that $B_1 = 0$ and $A_1 = 1$. Hence $f(z) = z^{1/4}$ and we recognize that this power function maps the upper half-plane onto the wedge $0 \leq \text{Arg } w \leq \pi/4$.
13. From (3), $f'(z) = A(z+1)^{(\alpha_1/\pi)-1} z^{(\alpha_2/\pi)-1} (z-1)^{(\alpha_3/\pi)-1}$. But as $u_1 \rightarrow 0$, $\alpha_1 \rightarrow \pi/2$, $\alpha_2 \rightarrow 2\pi$, and $\alpha_3 \rightarrow \pi/2$. This suggests that we examine

$$f'(z) = A(z+1)^{-1/2} z(z-1)^{-1/2} = A \frac{z}{(z^2-1)^{1/2}}.$$

Therefore, $f(z) = A(z^2-1)^{1/2} + B$. But $f(-1) = f(1) = 0$ and $f(0) = ai$. This implies that $B = 0$ and $ai = Ai$ or $A = a$. Therefore, $f(z) = a(z^2-1)^{1/2}$. By expressing $f(z)$ as the composite of $z_1 = z^2$, $z_2 = z_1 - 1$, $z_3 = z_2^{1/2}$ and $w = az_3$ we can show that the image of the upper half-plane is R' .

14. If $w(t) = u(t) + iv(t)$, then $w'(t) = u'(t) + iv'(t)$ and so

$$\tan(\arg w'(t)) = \frac{v'(t)}{u'(t)} = \frac{dv}{du}.$$

If $\arg w'(t)$ is constant, then $dv/du = m$ or $v = mu + b$ for some constants m and b .

EXERCISES 20.5

Poisson Integral Formulas

1. Using (3) with $x_0 = -1$, $x_1 = 0$, $x_2 = 1$ and $u_1 = -1$ and $u_2 = 1$,

$$u = -\frac{1}{\pi} \text{Arg} \left(\frac{z}{z+1} \right) + \frac{1}{\pi} \text{Arg} \left(\frac{z-1}{z} \right).$$

2. Using (3) with $x_0 = -2$, $x_1 = 0$, $x_2 = 1$ and $u_1 = 5$ and $u_2 = 1$,

$$u = \frac{5}{\pi} \text{Arg} \left(\frac{z}{z+2} \right) + \frac{1}{\pi} \text{Arg} \left(\frac{z-1}{z} \right).$$

3. The harmonic function

$$u_1 = \frac{5}{\pi} [\pi - \text{Arg}(z-1)] = \begin{cases} 5, & x > 1 \\ 0, & x < 1 \end{cases}, \quad \text{and} \quad u_2 = -\frac{1}{\pi} \text{Arg} \left(\frac{z+1}{z+2} \right) + \frac{1}{\pi} \text{Arg} \left(\frac{z}{z+1} \right)$$

from (3) satisfies all boundary conditions except that $u_2 = 0$ for $x > 1$. Therefore $u = u_1 + u_2$ is the solution to the given Dirichlet problem.

4. The harmonic function

$$u_1 = \frac{1}{\pi} \text{Arg}(z+2) = \begin{cases} 1, & x < -2 \\ 0, & x > -2 \end{cases}, \quad \text{and} \quad u_2 = -\frac{1}{\pi} \text{Arg} \left(\frac{z+1}{z+2} \right) + \frac{1}{\pi} \text{Arg} \left(\frac{z+1}{z-1} \right)$$

satisfies all boundary conditions except that $u_2 = 0$ for $x < -2$. Therefore $u = u_1 + u_2$ is the solution to the given Dirichlet problem.

5. By Theorem 20.5,

$$u(x, y) = \frac{y}{\pi} \int_0^1 \frac{t^2}{(x-t)^2 + y^2} dt.$$

If we let $s = x - t$, this integral can be expressed in terms of the natural log and inverse tangent. Using *Maple* we obtain

$$u = \frac{y}{\pi} \left(\frac{y^2 - x^2}{y} \left[\tan^{-1} \left(\frac{x-1}{y} \right) - \tan^{-1} \left(\frac{x}{y} \right) \right] + x \ln \frac{(x-1)^2 + y^2}{x^2 + y^2} + 1 \right).$$

6. From Theorem 20.5,

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\cos t}{(x-t)^2 + y^2} dt = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\cos(x-s)}{s^2 + y^2} ds$$

letting $s = x - t$. But $\cos(x-s) = \cos x \cos s + \sin x \sin s$. It follows that

$$u(x, y) = \frac{y \cos x}{\pi} \int_{-\infty}^{\infty} \frac{\cos s}{s^2 + y^2} ds + \frac{y \sin x}{\pi} \int_{-\infty}^{\infty} \frac{\sin s}{s^2 + y^2} ds = \frac{y \cos x}{\pi} \left(\frac{\pi e^{-y}}{y} \right) = e^{-y} \cos x, \quad y > 0.$$

7. The mapping $f(z) = z^2$ maps R onto the upper half-plane R' . The corresponding boundary value problem in R' is shown in the figure. From (3),

$$U = \frac{5}{\pi} \text{Arg}(w+1) + \frac{1}{\pi} \text{Arg} \left(\frac{w-1}{w} \right)$$

is the solution in R' . Therefore

$$u = U(f(z)) = \frac{5}{\pi} \text{Arg}(z^2 + 1) + \frac{1}{\pi} \text{Arg} \left(\frac{z^2 - 1}{z^2} \right)$$

is the solution to the original Dirichlet problem in R .

8. Using H-4 with $a = 3$, $f(z) = \cos(\pi z/3)$ maps R onto the upper half-plane R' . The corresponding Dirichlet problem in R' is shown in the figure. From (3),

$$U = \frac{1}{\pi} [\pi - \text{Arg}(w-1)] + \frac{1}{\pi} \text{Arg}(w+1)$$

is the solution in R' . Therefore

$$u = U(f(z)) = 1 + \frac{1}{\pi} [\text{Arg}(\cos(\pi z/3) + 1) - \text{Arg}(\cos(\pi z/3) - 1)]$$

is the solution to the original Dirichlet problem in R .

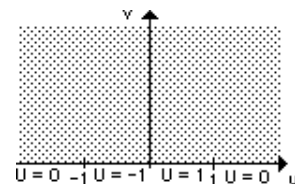
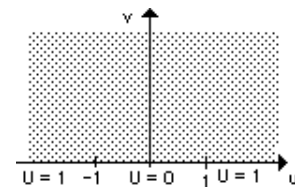
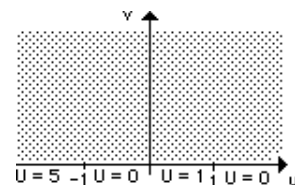
9. Using H-1, $f(z) = i \frac{1-z}{1+z}$ maps R onto the upper half-plane R' . The corresponding Dirichlet problem in R' is shown in the figure. From (3),

$$U = -\frac{1}{\pi} \text{Arg} \left(\frac{w}{w+1} \right) + \frac{1}{\pi} \text{Arg} \left(\frac{w-1}{w} \right) = \frac{1}{\pi} \left[\text{Arg} \left(\frac{w-1}{w} \right) - \text{Arg} \left(\frac{w}{w+1} \right) \right].$$

The harmonic function $u = U(f(z))$ may be simplified to

$$u = \frac{1}{\pi} \left[\text{Arg} \left(\frac{(1-i)z - (1+i)}{1-z} \right) - \text{Arg} \left(\frac{1-z}{-(1+i)z + 1-i} \right) \right]$$

and is the solution to the original Dirichlet problem in R .



20.5 Poisson Integral Formulas

10. Using H-5, $f(z) = \left(\frac{1+z}{1-z}\right)^2$ maps R onto the upper half-plane R' . The corresponding Dirichlet problem in R' is shown in the figure. From (3),

$$U = \frac{1}{\pi} \operatorname{Arg} \left(\frac{w}{w+1} \right) + \frac{1}{\pi} [\pi - \operatorname{Arg}(w-1)] = 1 - \frac{1}{\pi} \operatorname{Arg}(w-1) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{w}{w+1} \right).$$

The harmonic function $u = U(f(z))$ may be simplified to

$$u = 1 - \frac{1}{\pi} \operatorname{Arg} \left(\frac{4z}{(1-z)^2} \right) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{(1+z)^2}{2(1+z^2)} \right).$$

11. From Theorem 20.6, $u(x, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{t^2}{\pi^2} \frac{1-|z|^2}{|e^{it}-z|^2} dt$. Therefore,

$$u(0, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{t^2}{\pi^2} dt = \frac{1}{3}.$$

To estimate $u(0.5, 0)$ and $u(-0.5, 0)$ we must use a numerical integration method. With the aid of Simpson's Rule, $u(0.5, 0) = 0.1516$ and $u(-0.5, 0) = 0.5693$.

12. From Theorem 20.6, $u(x, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-|t|} \frac{1-|z|^2}{|e^{it}-z|^2} dt$. Therefore,

$$u(0, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-|t|} dt = \frac{1}{\pi} \int_0^{\pi} e^{-t} dt = \frac{1}{\pi} (1 - e^{-\pi}).$$

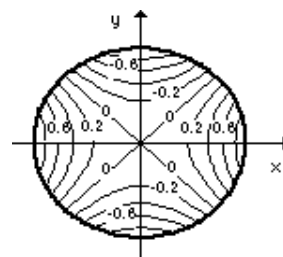
With the aid of Simpson's Rule, $u(0.5, 0) = 0.5128$ and $u(-0.5, 0) = 0.1623$.

13. $u(0, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(e^{it}) dt$. The latter integral is just the average value of $u(e^{it})$ for $-\pi \leq t \leq \pi$.

14. For $u(e^{i\theta}) = \cos 2\theta$, the Fourier series solution (6) reduces to

$$u(r, \theta) = r^2 \cos 2\theta = \operatorname{Re}(z^2) \quad \text{or} \quad u(x, y) = x^2 - y^2.$$

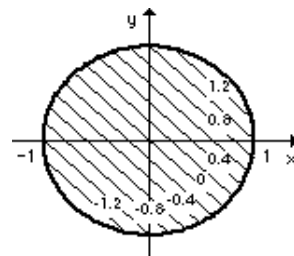
The corresponding system of level curves is shown in the figure.



15. For $u(e^{i\theta}) = \sin \theta + \cos \theta$, the Fourier series solution (6) reduces to

$$u(r, \theta) = r \sin \theta + r \cos \theta \quad \text{or} \quad u(x, y) = y + x.$$

The corresponding system of level curves is shown in the figure.



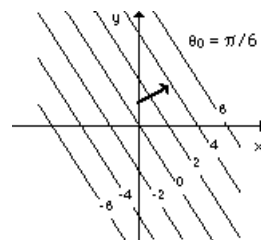
EXERCISES 20.6

Applications

1. $g(z) = \cos \theta_0 - i \sin \theta_0 = e^{-i\theta_0}$ is analytic everywhere and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = e^{-i\theta_0} z$ and

$$\phi(x, y) = \operatorname{Re}(G(z)) = x \cos \theta_0 + y \sin \theta_0.$$

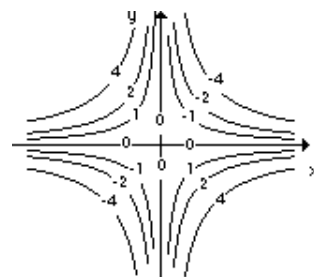
The equipotential lines (corresponding to $\theta_0 = \pi/6$) are shown in the figure.



2. $g(z) = -y + xi = i(x + iy) = iz$ is analytic everywhere and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = \frac{i}{2} z^2$ and

$$\phi(x, y) = \operatorname{Re}\left(\frac{i}{2} z^2\right) = -xy.$$

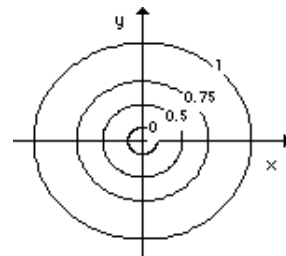
The equipotential lines $-xy = c$ are shown in the figure and are hyperbolas.



3. $g(z) = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2} i = \frac{1}{z}$ is analytic for $z \neq 0$ and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = \operatorname{Ln} z$ and

$$\phi(x, y) = \operatorname{Re}(G(z)) = \frac{1}{2} \log_e(x^2 + y^2).$$

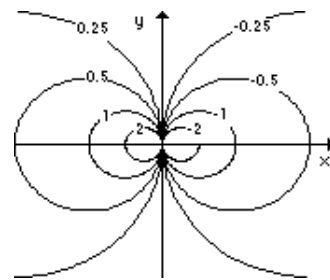
The equipotential lines $\phi(x, y) = c$ are circles $x^2 + y^2 = e^{2c}$ and are shown in the figure.



4. $g(z) = \frac{x^2 - y^2 - 2xyi}{(x^2 + y^2)^2} = \frac{1}{z^2}$ is analytic for $z \neq 0$ and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = -\frac{1}{z}$ and

$$\phi(x, y) = \operatorname{Re}\left(-\frac{1}{z}\right) = -\frac{x}{x^2 + y^2}.$$

The equipotential lines $-\frac{x}{x^2 + y^2} = c$ can be written as $\left(x + \frac{1}{2c}\right)^2 + y^2 = \left(\frac{1}{2c}\right)^2$ for $c \neq 0$. See the figure.



5. The mapping $f(z) = z^4$ maps the wedge $0 \leq \operatorname{Arg} z \leq \pi/4$ to the upper half-plane R' and $U = \frac{1}{\pi} \operatorname{Arg} w$ is the solution to the corresponding Dirichlet problem in R' . Therefore,

$$\phi(x, y) = U(z^4) = \frac{1}{\pi} \operatorname{Arg} z^4 = \frac{4}{\pi} \operatorname{Arg} z$$

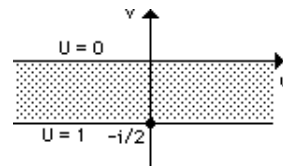
is the potential in the wedge. A complex potential is $G(z) = \frac{4}{\pi} \operatorname{Ln} z$ and, since $\phi(r, \theta) = \frac{4}{\pi} \theta$, the equipotential

20.6 Applications

lines are the rays $\theta = \frac{\pi}{4}c$. Finally

$$\mathbf{F} = \overline{G'(z)} = \frac{4}{\pi} \frac{1}{\bar{z}} = \frac{4}{\pi} \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right).$$

6. The function $f(z) = \frac{1}{z}$ maps the original region R to the strip $-\frac{1}{2} \leq v \leq 0$ (see Example 2, Section 20.1). The boundary conditions transfer as shown in the figure. $U = -2v$ is the solution in the horizontal strip and so



$$\phi(x, y) = -2\text{Im} \left(\frac{1}{z} \right) = \frac{2y}{x^2 + y^2}$$

is the potential in the original region R . The equipotential lines $\frac{2y}{x^2 + y^2} = c$ may be written as

$x^2 + \left(y + \frac{1}{c}\right)^2 = \left(\frac{1}{c}\right)^2$ for $c \neq 0$ and are circles. If $c = 0$, we obtain the line $y = 0$. Note that

$\phi(x, y) = \text{Re} \left(\frac{2i}{z} \right)$ and so $G(z) = \frac{2i}{z}$ is a complex potential. The corresponding vector field is

$$\mathbf{F} = \overline{G'(z)} = \frac{2i}{\bar{z}^2} = \left(\frac{-4xy}{(x^2 + y^2)^2}, \frac{2(x^2 - y^2)}{(x^2 + y^2)^2} \right).$$

7. Using H-5, $w = \left(\frac{1-z}{1+z}\right)^2 = \left(\frac{z-1}{z+1}\right)^2$ maps R onto the upper half-plane R' and $U = \frac{1}{\pi} \text{Arg } w$ is the solution to the corresponding Dirichlet problem in R' . Therefore,

$$\mu = \frac{1}{\pi} \text{Arg} \left(\frac{z-1}{z+1} \right)^2.$$

In R' the equipotential lines are rays $\theta = \theta_0$. The inverse transformation is $z = S(T(w))$ where $T(w) = w^{1/2}$ and $S(w) = \frac{1+w}{1-w}$. T maps the ray $\theta = \theta_0$ to the ray $\theta = \theta_0/2$ and S maps $\theta = \theta_0/2$ to an arc of a circle since $S(0) = 1$ and $S(\infty) = -1$ and S is a linear fractional transformation.

8. Using C-1 with $b = 2$ and $c = 4$, we have $a = \frac{3 + \sqrt{5}}{2}$ and $r_0 = \frac{7 - 3\sqrt{5}}{2}$, so $T(z) = \frac{z-a}{az-1}$ maps R onto the annular region R' defined by $r_0 \leq |w| \leq 1$. The solution to the corresponding Dirichlet problem in R' is $U = \frac{\log_e |w|}{\log_e r_0}$ and so

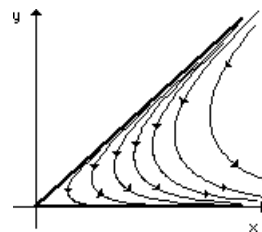
$$\phi = \frac{1}{\log_e r_0} \log_e \left| \frac{z-a}{az-1} \right|.$$

The level curves of U are circles $|w| = r$ where $r_0 < r < 1$, and the equipotential lines $\phi(x, y) = c$ are the images of these circles under the inverse transformation $T^{-1}(w) = \frac{-w+a}{-aw+1}$. T^{-1} has a pole at $w = \frac{1}{a} = \frac{2}{3 + \sqrt{5}}$ (≈ 0.38) and $r_0 < \frac{1}{a} < 1$. All circles $|w| = r$ are mapped onto circles in the z -plane with the exception of $|w| = \frac{1}{a}$ which is mapped onto a line.

9. (a) $\psi(x, y) = \text{Im}(z^4) = 4xy(x^2 - y^2)$ and so $\psi(x, y) = 0$ when $y = x$ and $y = 0$.

(b) $\mathbf{V} = \overline{G'(z)} = \overline{4z^3} = 4(x^3 - 3xy^2, y^3 - 3x^2y)$

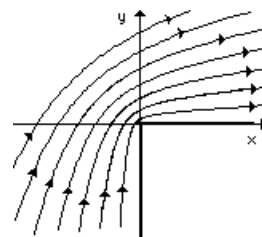
- (c) In polar coordinates $r^4 \sin 4\theta = c$ or $r = (c \csc 4\theta)^{1/4}$, for $0 < \theta < \pi/4$, are the streamlines. See the figure.



10. (a) Since $G(re^{i\theta}) = r^{2/3}e^{i2\theta/3}$, $\psi(r, \theta) = \text{Im}(G(re^{i\theta})) = r^{2/3} \sin \frac{2\theta}{3}$. Note that $\psi = 0$ on the boundary where $\theta = 0$ and $\theta = 3\pi/2$.

(b) $\mathbf{V} = \overline{G'(z)} = \frac{2}{3}z^{-1/3}$. Therefore, letting $z = re^{i\theta}$, $\mathbf{V} = \frac{2}{3}r^{-1/3}(\cos(\theta/3), \sin(\theta/3))$ for $0 < \theta < 3\pi/2$.

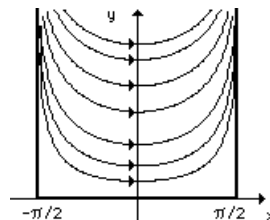
- (c) $r^{2/3} \sin(2\theta/3) = c$ implies that $r = [c \csc(2\theta/3)]^{3/2}$ for $0 < \theta < 3\pi/2$. The streamlines are shown in the figure.



11. (a) $\psi(x, y) = \text{Im}(\sin z) = \cos x \sinh y$ and $\psi(x, y) = 0$ when $x = \pm\pi/2$ or when $y = 0$.

(b) $\mathbf{V} = \overline{G'(z)} = \overline{\cos z} = (\cos x \cosh y, \sin x \sinh y)$.

- (c) $\cos x \sinh y = c \implies y = \sinh^{-1}(c \sec x)$ and the streamlines are shown in the figure.



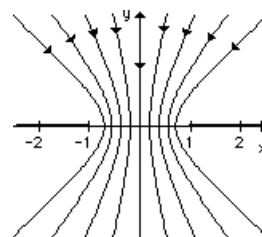
12. (a) The image of R under $w = i \sin^{-1} z$ is the horizontal strip (see E-6) $-\pi/2 \leq v \leq \pi/2$ and

$$\psi(x, y) = \text{Im}(i \sin^{-1} z) = \begin{cases} \pi/2, & x \geq 1 \\ -\pi/2, & x \leq -1 \end{cases}.$$

Each piece of boundary is therefore a streamline.

(b) $\mathbf{V} = \overline{G'(z)} = \frac{i}{(1-z^2)^{1/2}} = \frac{-i}{(1-\bar{z}^2)^{1/2}}$

- (c) The streamlines are the images of the lines $v = b$, $-\pi/2 < b < \pi/2$ under $z = -i \sin w$ and are therefore hyperbolas. See Example 2, Section 20.2, and the figure. Note that at $z = 0$, $v = -i$ and the flow is downward.



13. (a) If $z = re^{i\theta}$, $G(re^{i\theta}) = r^2 e^{2i\theta} + \frac{1}{r^2} e^{-2i\theta}$ and so

$$\psi(r, \theta) = \text{Im}(G(re^{i\theta})) = \left(r^2 - \frac{1}{r^2}\right) \sin 2\theta.$$

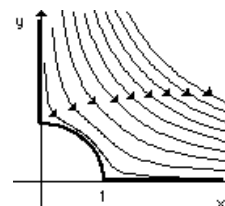
Note that $\psi = 0$ when $r = 1$ or when either $\theta = 0$ or $\theta = \pi/2$. Therefore $\psi = 0$ on the boundary of R .

- (b) $\mathbf{V} = \overline{G'(z)} = \overline{(2z - 2z^{-3})}$ and so in polar coordinates

$$\mathbf{V} = 2re^{-i\theta} - \frac{2}{r^3} e^{3i\theta} = 2(r \cos \theta - r^{-3} \cos 3\theta, -r \sin \theta - r^{-3} \sin 3\theta).$$

- (c) In rectangular coordinates, the streamlines are

$$\psi(x, y) = 2xy \left[1 - \frac{1}{(x^2 + y^2)^2}\right] = c.$$

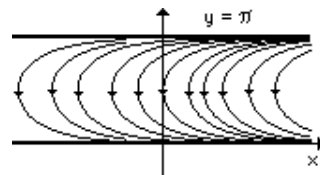


20.6 Applications

14. (a) $\psi(x, y) = \operatorname{Im}(e^z) = e^x \sin y$ and so $\psi = 0$ when $y = 0$ or π . Therefore $\psi = 0$ on the boundary of R .

(b) $\mathbf{V} = \overline{G'(z)} = \overline{e^z} = (e^x \cos y - e^x \sin y)$

(c) The streamlines are $e^x \sin y = c$ and so $x = \log_e(c \csc y)$ for $0 < y < \pi$. See the figure.

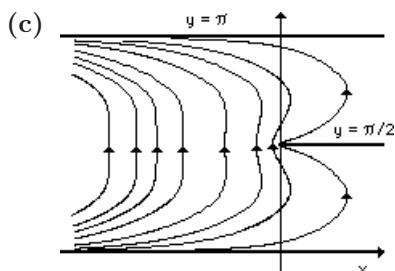


15. (a) For $f(z) = \pi i - \frac{1}{2}[\operatorname{Ln}(z+1) + \operatorname{Ln}(z-1)]$

$$f(t) = \pi i - \frac{1}{2}[\log_e |t+1| + \log_e |t-1| + i\operatorname{Arg}(t+1) + i\operatorname{Arg}(t-1)]$$

and so $\operatorname{Im}(f(t)) = \begin{cases} 0, & t < -1 \\ \pi/2, & -1 < t < 1 \\ \pi, & t > 1 \end{cases}$. Hence $\operatorname{Im}(G(z)) = \psi(x, y) = 0$ on the boundary of R .

(b) $x = -\frac{1}{2}[\log_e |t+1+ic| + \log_e |t-1+ic|]$, $y = \pi - \frac{1}{2}[\operatorname{Arg}(t+1+ic) + \operatorname{Arg}(t-1+ic)]$ for $c > 0$

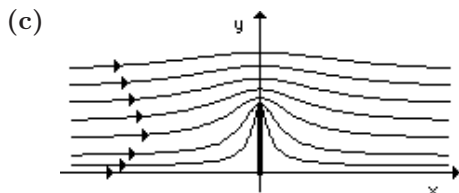


16. (a) For $f(z) = (z^2 - 1)^{1/2}$,

$$f(t) = |t^2 - 1|^{1/2} \cos\left(\frac{1}{2}\operatorname{Arg}(t^2 - 1)\right) + i|t^2 - 1|^{1/2} \sin\left(\frac{1}{2}\operatorname{Arg}(t^2 - 1)\right)$$

and so $f(t) = \begin{cases} |t^2 - 1|^{1/2}, & |t| > 1 \\ i|t^2 - 1|^{1/2}, & |t| < 1 \end{cases}$. Hence $\operatorname{Im}(G(z)) = 0$ on the boundary of R .

(b) $x = |(t+ic)^2 - 1|^{1/2} \cos\left(\frac{1}{2}\operatorname{Arg}((t+ic)^2 - 1)\right)$, $y = |(t+ic)^2 - 1|^{1/2} \sin\left(\frac{1}{2}\operatorname{Arg}((t+ic)^2 - 1)\right)$ for $c > 0$



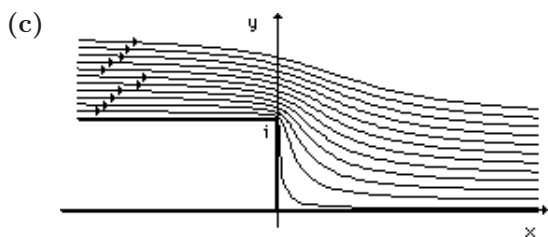
17. (a) For $f(z) = \frac{1}{\pi}[(z^2 - 1)^{1/2} + \cosh^{-1} z]$,

$$f(t) = \frac{1}{\pi}[(t^2 - 1)^{1/2} + \cosh^{-1} t] = \frac{1}{\pi}[(t^2 - 1)^{1/2} + \operatorname{Ln}(t + (t^2 - 1)^{1/2})]$$

and so $\operatorname{Im}(f(t)) = \begin{cases} 1, & t < -1 \\ 0, & t > 1 \end{cases}$ and $\operatorname{Re}(f(t)) = 0$, for $-1 < t < 1$. Hence $\operatorname{Im}(G(z)) = \psi(x, y) = 0$ on the boundary of R .

(b) $x = \operatorname{Re}\left(\frac{1}{\pi}\left[(t+ic)^2 - 1\right]^{1/2} + \cosh^{-1}(t+ic)\right)$,

$y = \operatorname{Im}\left(\frac{1}{\pi}\left[(t+ic)^2 - 1\right]^{1/2} + \cosh^{-1}(t+ic)\right)$ for $c > 0$



18. (a) For $f(z) = 2(z+1)^{1/2} + \text{Ln} \left(\frac{(z+1)^{1/2} - 1}{(z+1)^{1/2} + 1} \right)$,

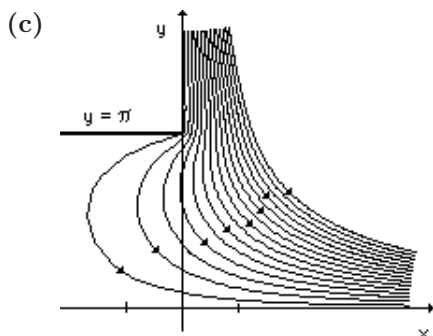
$$f(t) = 2(t+1)^{1/2} + \text{Ln} \frac{(t+1)^{1/2} - 1}{(t+1)^{1/2} + 1}.$$

If we write $(t+1)^{1/2} = |t+1|^{1/2} e^{(i/2)\text{Arg}(t+1)}$, we may conclude that

$$\text{Im}(f(t)) = \begin{cases} 0, & t > 0 \\ \pi, & -1 < t < 0 \end{cases} \quad \text{and} \quad \text{Re}(f(t)) = 0 \text{ for } t < -1.$$

Therefore $\text{Im}(G(z)) = \psi(x, y) = 0$ on the boundary of R .

(b) $x = \text{Re} \left[2(t+ic+1)^{1/2} + \text{Ln} \frac{(t+ic+1)^{1/2} - 1}{(t+ic+1)^{1/2} + 1} \right]$
 $y = \text{Im} \left[2(t+ic+1)^{1/2} + \text{Ln} \frac{(t+ic+1)^{1/2} - 1}{(t+ic+1)^{1/2} + 1} \right] \quad \text{for } c > 0$



19. In Example 5, $\mathbf{V} = (2x, -2y)$ and so the only stagnation point occurs at $z = 0$. In Example 6, $\mathbf{V} = 1 - 1/\bar{z}^2$ and so if $\mathbf{V} = \mathbf{0}$, $\bar{z}^2 = 1$. Therefore $z = 1, -1$ are the only stagnation points.

20. (a) $\psi(x, y) = \text{Im}(G(z)) = k \text{Arg}(z - x_1)$ and so if $\psi(x, y) = c$, $\text{Arg}(z - x_1)$ is constant. This implies that the streamlines are rays with vertex at $z = x_1$.

(b) $\mathbf{V} = \overline{G'(z)} = \frac{\overline{k}}{z - x_1} = \frac{k}{\overline{z - x_1}} = \frac{k}{|z - x_1|^2} (z - x_1).$

The direction of the flow is determined by the sign of k , and if $k < 0$ the flow is directed towards $z = x_1$.

21. $f(z) = z^2$ maps the first quadrant onto the upper half-plane and $f(\xi_0) = f(1) = 1$. Therefore $G(z) = \text{Ln}(z^2 - 1)$ is the complex potential, and so

$$\psi(x, y) = \text{Arg}(z^2 - 1) = \tan^{-1} \left(\frac{2xy}{x^2 - y^2 - 1} \right)$$

is the streamline function where $\tan^{-1}$ is chosen to be between 0 and π . If $\psi(x, y) = c$, then $x^2 + Bxy - y^2 - 1 = 0$ where $B = -2 \cot c$. Each hyperbola in the family passes through $(1, 0)$ and the boundary of the first quadrant satisfies $\psi(x, y) = 0$.

20.6 Applications

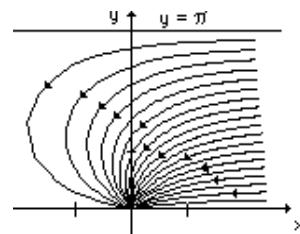
22. (a) From E-5, $f(z) = e^z$ maps the horizontal strip $0 < y < \pi$ onto the upper half-plane and $f(\xi_0) = f(0) = 1$. Therefore $G(z) = k \operatorname{Ln}(e^z - 1)$ is a complex potential. To construct a sink at $\xi_0 = 0$, we must have $k < 0$.

(b) $\psi(x, y) = \operatorname{Im}(k \operatorname{Ln}(e^z - 1)) = k \operatorname{Arg}(e^z - 1) = k \tan^{-1} \left(\frac{e^x \sin y}{e^x \cos y - 1} \right)$

where $\tan^{-1}$ is chosen to be between 0 and π . If we set $k = -1$, then the streamlines $\psi(x, y) = c$, $-\pi < c < 0$, satisfy $e^x [B \cos y - \sin y] = B$ where $B = -\tan c$, and so

$$x = \log_e \left[\frac{B}{B \cos y - \sin y} \right].$$

Note that B will vary over all real values. The streamlines are also the images of rays through $w = 1$ under the inverse transformation $z = \operatorname{Ln} w$. See the figure.



23. $\psi = \operatorname{Im}(G(z)) = k \operatorname{Arg}(z - 1) - k \operatorname{Arg}(z + 1) = k \operatorname{Arg} \left(\frac{z - 1}{z + 1} \right)$.

(See (1) and (2) in this section in the text). In rectangular coordinates

$$\psi(x, y) = k \tan^{-1} \left(\frac{2y}{x^2 + y^2 - 1} \right)$$

where $\tan^{-1}$ is chosen to be between 0 and π . Level curves $\psi(x, y) = c$ can be put in the form

$$x^2 + y^2 - 2By = 1 \quad \text{or} \quad x^2 + (y - B)^2 = 1 + B^2.$$

Each member of the family passes through $(1, 0)$ and $(-1, 0)$.

24. (a) $\mathbf{V} = \frac{a + ib}{\bar{z}} = \left(\frac{ax - by}{x^2 + y^2}, \frac{bx + ay}{x^2 + y^2} \right)$ and since $(x'(t), y'(t)) = \mathbf{V}$, the path of the particle satisfies

$$\frac{dx}{dt} = \frac{ax - by}{x^2 + y^2}, \quad \frac{dy}{dt} = \frac{bx + ay}{x^2 + y^2}.$$

(b) Switching to polar coordinates,

$$\begin{aligned} \frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} \left(\frac{ax^2 - bxy}{r^2} + \frac{bxy + ay^2}{r^2} \right) = \frac{a}{r} \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} \left(\frac{-axy + by^2}{r^2} + \frac{bx^2 + axy}{r^2} \right) = \frac{b}{r^2}. \end{aligned}$$

Therefore $\frac{dr}{d\theta} = \frac{a}{b} r$ and so $r = ce^{a\theta/b}$.

- (c) $\frac{dr}{dt} = \frac{a}{r} < 0$ if and only if $a < 0$, and in this case r is decreasing and the curve spirals inward. $\frac{d\theta}{dt} = \frac{b}{r^2} < 0$ if and only if $b < 0$, and in this case θ is decreasing and the curve is traversed clockwise.

CHAPTER 20 REVIEW EXERCISES

1. $f(z) = x^2 - y^2 + 2xyi$ and so the hyperbola $xy = 2$ is mapped onto the line $v = 4$.
2. $-i = e^{-i\pi/2}$ and so $f(z) = -iz$ is a rotation through -90° .
3. The wedge $0 \leq \text{Arg } w \leq 2\pi/3$. See figure 20.6 in the text.
4. $f'(z) = \sinh z = 0$ for $z = \pm\pi i$, and so $f(z) = \cosh z$ is conformal except at $z = \pm n\pi i$.
5. True, by Theorem 20.2. $\text{Arg } w$ is harmonic and is the upper half-plane $v \geq 0$.
6. A line, since $|z - 1| = 1$ passes through the pole at $z = 2$.
7. $0, 1, \infty$.
8. True. $\alpha_1 = \alpha_2 = \alpha_3 = \pi/2$
9. False. $\overline{g(z)} = P - iQ$ is analytic. See Theorem 20.7.
10. $iG(z) = -\psi(x, y) + i\phi(x, y)$ is analytic in R and $\text{Im}(G(z)) = \phi(x, y)$. True
11. If $z = re^{i\theta}$ and $0 < \theta < \pi/2$, $w = \text{Ln } z = \log_e r + i\theta$. Therefore $v = \theta$ and so $0 < v < \pi/2$. The image of the first quadrant is the strip $0 < v < \pi/2$ in the w -plane. Rays $\theta = \theta_0$ are mapped onto horizontal lines $v = \theta_0$.
12. First use $z_1 = z^2$ to map the first quadrant onto the upper half-plane $y_1 \geq 0$, and segment AB to the negative real axis. We then use $w = \frac{1}{\pi}(z_1 + \text{Ln } z_1 + 1)$ to map this half-plane onto the target region R' . The composite transformation is

$$w = \frac{1}{\pi}[z^2 + \text{Ln}(z^2) + 1]$$

and the image of AB is the ray extending to the left from $w = i$ along the line $v = 1$.

13. From H-4, $z_1 = \cos \pi z$ maps R onto the upper half-plane $y_1 \geq 0$, and AB onto the real segment $(-\infty, -1]$. Using C-4, $w = \frac{i - z_1}{1 + z_1}$ maps this upper half-plane onto the target region R' . The composite transformation is

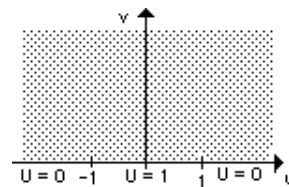
$$w = \frac{i - \cos \pi z}{i + \cos \pi z}$$

and the image of AB is the circular arc lying in quadrants II and III.

14. The power function $w = z^4$ maps R onto the upper half-plane $v \geq 0$; the transferred boundary conditions are shown in the figure. From (2) of Section 20.5, $U = \frac{1}{\pi} \text{Arg} \left(\frac{w-1}{w+1} \right)$ and so

$$u = \frac{1}{\pi} \text{Arg} \left(\frac{z^4 - 1}{z^4 + 1} \right)$$

is the solution to the original Dirichlet problem in R .

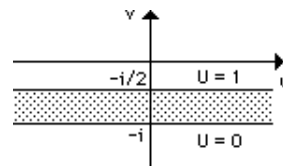


CHAPTER 20 REVIEW EXERCISES

15. The inversion $w = 1/z$ maps R onto the horizontal strip $-1 \leq v \leq -1/2$ and the transferred boundary conditions are shown in the figure. The solution in the strip is $U = 2v + 2$ and so

$$u = U\left(\frac{1}{z}\right) = 2\operatorname{Im}\left(\frac{1}{z}\right) + 2 = \frac{-2y}{x^2 + y^2} + 2$$

is the solution to the original Dirichlet problem in R .



16. Using the cross ratio formula,

$$\frac{(w-i)(-i+1)}{(w+1)(-2i)} = \frac{(z-1) \cdot 1}{1 \cdot (-2)}$$

we see that $w = -\frac{z-i}{z+i}$ maps $1, -1, \infty$ to $i, -i, -1$.

17. (a) We note that as $u_1 \rightarrow +\infty$, $\alpha_1 \rightarrow 0$, $\alpha_2 \rightarrow 2\pi$, and $\alpha_3 \rightarrow 0$.

(b) Since

$$f'(z) = A(z+1)^{(\alpha_1/\pi)-1} z^{(\alpha_2/\pi)-1} (z-1)^{(\alpha_3/\pi)-1},$$

this suggests that we examine

$$f'(z) = A(z+1)^{-1} z(z-1)^{-1} = A \frac{z}{z^2 - 1}.$$

We may write

$$f'(z) = \frac{A}{2} \left(\frac{1}{z+1} + \frac{1}{z-1} \right)$$

and so

$$f(z) = \frac{1}{2} A [\operatorname{Ln}(z+1) + \operatorname{Ln}(z-1)] + B.$$

- (c) For t real, we can write

$$f(t) = \frac{1}{2} A [\log_e |t+1| + \log_e |t-1| + i\operatorname{Arg}(t+1) + i\operatorname{Arg}(t-1)] + B.$$

Since $f(0) = \frac{\pi i}{2}$, $\frac{\pi i}{2} = \frac{\pi i}{2} A + B$. For t real,

$$f(t) = \begin{cases} \frac{1}{2} A [\ln |t^2 - 1| + 2\pi i] + B, & t < -1 \\ \frac{1}{2} A \ln |t^2 - 1| + B, & t > 1 \end{cases}.$$

If A is real and we require that $\operatorname{Im}(f(t)) = 0$ for $t < -1$, then $0 = A\pi i + \operatorname{Im}(B)$. If $\operatorname{Im}(f(t)) = \pi$ for $t > 1$, then $\pi = \operatorname{Im}(B)$. All three equations are satisfied by letting $B = \pi i$ and $A = -1$. Therefore

$$f(z) = \pi i - \frac{1}{2} [\operatorname{Ln}(z+1) + \operatorname{Ln}(z-1)].$$

18. (a) From Theorem 20.5,

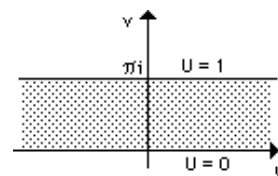
$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\sin t}{(x-t)^2 + y^2} dt = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\sin(x-s)}{s^2 + y^2} ds \quad (\text{letting } s = x-t).$$

But $\sin(x-s) = \sin x \cos s - \cos x \sin s$. We now proceed as in the solution to Problem 6, Section 20.5 to show that $u(x, y) = e^{-y} \sin x$.

- (b) For $u(e^{i\theta}) = \sin \theta$, the Fourier Series solution (6) in Section 20.5 reduces to $u(r, \theta) = r \sin \theta$.

19. If $f(w) = w + e^w + 1$, $G(z) = f^{-1}(z)$ maps R onto the strip $0 \leq v \leq \pi$ and the transferred boundary conditions are shown in the figure to the right. The solution for the strip is $U = v/\pi$ and so the solution in R is

$$\phi(x, y) = U(G(z)) = \frac{1}{\pi} \operatorname{Im}(G(z)) = \frac{1}{\pi} \psi(x, y).$$



Therefore the equipotential lines $\phi(x, y) = c$ are the streamlines $\psi(x, y) = c\pi$ of the flow in Figure 20.72.

20. $G(re^{i\theta}) = -r^{1/2} \sin(\theta/2) + i[r^{1/2} \cos(\theta/2) - 1]$ and so $\psi(r, \theta) = \sqrt{r} \cos(\theta/2) = 1$. If $\psi(r, \theta) = 0$, $r \cos^2(\theta/2) = 1$ or $r \cos \theta + r = 2$ (using $\cos^2(\theta/2) = (1 + \cos \theta)/2$.) In rectangular coordinates, $x + \sqrt{x^2 + y^2} = 2$ or $y^2 = 4 - 4x$. Therefore the boundary of R is a streamline. To sketch the streamlines, note that $\psi(r, \theta) = c$ implies that $r = (c + 1)^2 \sec^2(\theta/2)$. See the figure to the right.

